VOLCANOES

GOO

1. Volcanoes & Causes of Volcanic Eruptions 🌋💥

Definitions:

- **Endogenic Forces:** Forces relating to or arising from the interior of the earth (diastrophic and sudden movements). 🌍
- **Volcano:** A vent in the earth's crust from which fiery hot magma erupts from the earth's interior. 🕳️🔥

Key Points (Exhaustive Detail):

- Volcanic eruptions are sudden movements associated strictly with internal forces of the earth.
- These sudden movements bring about instant changes on the surface of the earth. ⚡
- **Cause 1: Heat and Pressure inside the Earth:** Temperature and pressure both increase from the surface towards the centre. Rocks are bad conductors of heat, so the earth's heat does not escape on its own. It melts rocks and builds up great pressure. This pressure forces the heat to find an escape route through fissures and cracks. 🌡️🧲
- **Cause 2: Plate Tectonics:** Most eruptions take place near plate margins. When movable plates slide past each other or collide due to intense heat, their edges produce faults along the lines of weakness. It is along these plate boundaries that the crust is weakened and hot magma gushes out. (The exact same forces that cause earthquakes cause volcanic eruptions). 🗺️💥
- **Cause 3: Magma Chamber:** Molten material under the crust melts weak rocks and creates a huge chamber. Fresh magma (silicate material, gases, water vapours) pours inside. Pressure acts vertically upwards. Once a route is found, it rises to the surface, which is why many volcanoes are found along plate boundaries. 🌋

The "Why?" Section ❓

- **Why do solid rock fragments (pyroclasts) erupt if they don't come from the magma chamber?** The rock fragments that a volcano erupts do not originate from the magma chamber itself; they are actively picked up by the magma in the course of its violent upward movement through the crust! 🪨⬆️

Things to Remember 🧠

Feature	Detail
Volcanic Vent	The conduit/pipe that joins the huge magma chamber reservoir to the earth's surface. 🚇
Line of Weakness	Plate margins and boundaries where the crust is weakened, producing faults for magma to escape. 〰️

2. Products & Structure of a Volcano 🧿🪨

Definitions:

- **Magma:** The molten rock material inside the earth. 🌋
- **Lava:** The molten rock material when it reaches the surface of the earth. 🔥
- **Pyroclasts:** The solid fragments and rock pieces ejected during an eruption. 🪨
- **Dust:** Fine solid particles. 🌬️
- **Lapilli:** Small stone-sized solid particles. 🪨
- **Bombs:** Large materials ejected during violent eruptions. 💣
- **Vent:** The opening (usually a circular pipe) in the earth's crust through which lava flows out. 🕳️

- **Volcanic Cone:** The typical cone-like shape formed by the accumulation of erupted fluid material around the vent. 🏞️

Key Points (Exhaustive Detail & Handwritten Notes!):

- A volcano erupts steam, gases, molten rock, dust, ashes, and other liquid and solid matter.
- Lava consists of many materials and gases. Steam is the most abundant gas that comes out.
- When steam is ejected in huge quantities, it forms clouds in the atmosphere, resulting in rainfall. 🌧️
- Other gases ejected include hydrogen sulphide, sulphur dioxide, carbon dioxide, and hydrogen. 💨
- There may be only one opening at the summit (vent). However, subsidiary vents are formed during repeated eruptions.
- The erupted fluid material collects around the vent, cools gradually *layer by layer*, and solidifies to give the volcano its typical cone-like shape. 🤩 🌋

3. Types of Volcanoes (Based on Frequency of Eruption) ⏱️ 🌋

(Note: The Page-Shift Protocol was engaged here to seamlessly stitch the "Dormant Volcanoes" section across pages 67 and 68!) 🧩 📖

Key Points:

- **1. Active Volcanoes:** Presently in an active state and have erupted in the recent past.
    - There are ~500 known active volcanoes. About 20 may erupt at any time. A few are in a state of permanent eruption.
    - *Critical Examples:* Mt. Stromboli and Mt. Etna (Italy); Pinatubo (Philippines); and Mauna Loa in Hawaii. 📌

- **2. Dormant Volcanoes:** Have not erupted in the recent historical period. They are regarded as 'sleeping volcanoes' and may become active at any time.
    - *Critical Examples:* Mt. Kilimanjaro (Africa). Mt. Vesuvius, though regarded as active, is actually rated as dormant because there has been no major eruption since 1944. Many other volcanoes in Philippines and Japan are examples. 📌
- **3. Extinct Volcanoes:** These have not erupted in the present geological period and are not likely to become active again.
    - *Critical Examples:* Mt. Aconcagua (South America) and Mt. Kenya (Africa). 📌

Table of Differences 🆚

Type	Eruption Frequency	Critical Examples 📌
Active	Erupted in recent past; currently active.	Mt. Etna, Pinatubo, Mauna Loa.
Dormant	Sleeping; not erupted in recent history but could.	Mt. Kilimanjaro, Mt. Vesuvius.
Extinct	Not erupted in present geological period; unlikely to erupt again.	Mt. Aconcagua, Mt. Kenya.

4. Effects of Volcanoes (Constructive Effects - Part 1) 🏗️🌍

Key Points:

- Volcanic eruptions are a sudden force but are also termed as 'Constructive forces' because they have a number of constructive effects on the surface of the earth. 🛠️

- Numerous types of landforms are created due to the cooling and solidification of magma below the earth's surface (intrusive) and lava on the surface (Extrusive).
- **(a) Volcanic Plateaus:** Derived from lava flows from volcanic eruptions. *Examples:* Plateau of Peninsular India (western Deccan), South African plateau, Columbian Plateau, and Ethiopian Plateau. 🌋🏞️
- **(b) Volcanic Mountains:** Built from material ejected from fissures. They are the most diverse because there are massive differences in volcanic eruptions as well as the materials they throw up.
    - *Example 1:* Mauna Loa in Hawaii rises 9 km from the ocean floor! 🌊📈
    - *Example 2:* Volcanic mountains are common in the Circum Pacific Belt (Mt. Fuji/Japan, Mt. Mayon/Philippines, Mt. Merapi/Sumatra, Mt. Agung/Bali, Mt. Cotopaxi/Ecuador). 📌
- **(c) Volcanic Plains:** These are vast and smooth plains formed by extensive volcanic flooding from volcanic centres.
    - *Example:* The Western Victorian Plains in Victoria, Australia. These are the finest examples, formed over the last six million years with the most recent eruption being... *(Text continues on next unseen page)* ⏳

Things to Remember (DO YOU KNOW Fact Box!) 🧠

- **Hayli Gubbi:** A shield volcano in the Afar Basin of Ethiopia on the East African Rift (EAR) zone. 🌍
- **Tectonic Event:** The Nubian, Somalian, and Arabian plates are actively diverging here.
- **Eruption Event:** It erupted unexpectedly on November 23, 2025, after 12,000 years (first time since the end of the last ice age). 💥
- **Impact:** It produced thick ash plumes 15 km high that drifted up to China, covering Oman, Yemen, India, Pakistan, and Southeast Asia. It released 220,000 tons of sulfur dioxide! 🌫️💨

4. Effects of Volcanoes (Constructive Effects - Part 2) 🏗️🌍

(Note: Stitched seamlessly from the previous page cutoff!) 🧩📖

Key Points:

- ...most recent eruption being at Mount Napier, 7200 years ago. Over 400 volcanic sites have been found in this region. 📌
- **(d) Caldera Lake:** During repeated eruptions, the summit of a volcano may be blown up. In its place, a large depression called a *caldera* is formed. These are generally formed when the magma chamber is no longer able to emit sufficient magma, which results in the collapse of a cone either partly or wholly. This *depression* eventually gets filled with water and forms a lake.
    - *Example:* Crater Lake in the USA is a Caldera lake. It is about 10km wide and is the deepest lake in the USA. 📌
- **(e) Hot Springs:** The movement of magma in the interior of the earth heats up underground water, particularly around magma chambers. When water turns into steam, it gushes up to the surface.
    - *Properties:* Hot springs (also known as Thermal Springs) contain many dissolved minerals. Since hot springs are a rich source of sulphur, they are considered good for health, especially for curing certain skin diseases.
    - *Example:* Sulphur hot springs at Manikaran in Kullu valley of Himachal Pradesh are a great tourist attraction. 📌
- **(f) Geysers:** A geyser is a fountain of hot water and steam that escapes when underground water comes into contact with hot volcanic rocks. ⛲
    - *Mechanism (Handwritten Note!):* This escape is driven directly by intense underground **"pressure"**. ✍️🗜️
    - *Example:* The 'Old Faithful' Geyser in Yellowstone National Park in Wyoming, USA. During an eruption, its water temperature at the vent has been measured at 95.6°C! 📌🌡️

- **2. Other Constructive Effects:**
    - **(a)** The lava and ash deposited during an eruption breaks down to provide valuable nutrients for the soil, thus creating fertile soil which is good for agriculture. 🌱
    - **(b)** The places that have a high level of heat due to volcanic activity inside the earth have a higher potential for generating geothermal energy. ⚡
    - **(c)** The dramatic scenery created by volcanic eruptions attracts tourists, bringing income to the area. 📸

The "Why?" Section ❓

- **Why do Calderas form?** A caldera forms because the underlying magma chamber empties out and can no longer support the weight of the volcanic cone above it, causing the entire summit to collapse inward into a massive depression! 🕳️💥

5. Destructive Effects of Volcanoes 🌋💥

Key Points:

- **(i) Destruction of Life and Property:** When volcanoes erupt, they destroy life and property. The hot lava, together with ash and dust, destroy not only human beings but also animal and plant life. ☠️🏚️
- **(ii) Agricultural Ruin:** Agricultural fields are completely covered with thick layers of volcanic ash and dust, making them entirely unsuitable for cultivation. 🌾🚫
- **(iii) Floods and Landslides:** Huge clouds are formed immediately after the eruption of volcanoes, causing heavy rains which result directly in massive floods and landslides. ⛈️🌊
- **(iv) Pollution:** Volcanoes also emit poisonous gases which pollute the environment and cause severe health problems. 💨😷

- **(v) Tsunamis:** Explosive volcanoes occurring in ocean islands are followed by high Tsunami waves. They flood the area and destroy property, people, animals, and crops. 🌊💥 *(Checkmarks noted on all points!)*

6. Important Volcanic Zones of the World 🗺️🌋

Definitions:

- **Circum Pacific Belt (Pacific Ring of Fire):** A major zone of volcanoes surrounding the Pacific Ocean, known for intense tectonic and volcanic activity. ⭕🔥
- **Mid-world Mountain Belt:** A volcanic zone stretching across Europe and Asia, aligning with major mountain ranges. 🏔️

Key Points:

- **Zone 1: Circum Pacific Belt:**
  - The volcanoes occur near Earthquake belts around the young fold mountains. 🏔️
  - It is called the Pacific Ring of Fire because *over 80%* of the total number of active volcanoes are concentrated in this specific region! 📈
  - *Stats:* Among these are 40 active volcanoes in the Andes, over 100 in the Philippines, 70 in Indonesia, and 40 in Japan. 📊
- **Zone 2: Mid-world Mountain Belt:**
  - *Critical Examples:* Mt. Vesuvius, Mt. Etna, and Mt. Stromboli (the famous active volcanoes) are located in this belt. 📌
  - The *only* active volcano in India is located in the Andaman and Nicobar Islands, which falls exactly in this belt! IN
  - The Hawaiian Islands, Canary Islands, and Mt. Kenya are all located in the Mid-world Mountain Belt. 📌